

Dark Matter: Crash Course Astronomy #41

- Measure Doppler shift --> calculate velocity
- Velocities flatten out at greater distances from the center of a galaxy instead of declining
- Gravity is constant throughout the disk of a galaxy
- Rubin found that there is 5-6 times more dark matter than visible matter in galaxies
- Axions--> never been detected, but properties match that of dark matter:
 - Have mass, so could affect the gravity of galaxies
 - Don't emit enough light to be detectable
 - Don't interact well with visible matter
- When clusters collide, they don't pass through one another as typical galaxies do. Rather, the large amount of hot gas in between them collides and causes x-ray emissions
- Dark matter distorts the images of galaxies behind it, and that distortion can be measured to calculate the amount of dark matter
- Larger structures observed in the universe couldn't have aggregated without the influence of dark matter
- ★ How is gravitational warping of light detected by gravitational lensing techniques?
- ★ If axions are only one theory for dark matter composition, what are the other leading theories?
- ★ How is the Doppler shift measured?
- ★ Laser Doppler Vibrometry
- ★ What are the steps to mapping the structure of dark matter?

The Absurd Search for Dark Matter

- Dark matter is thought to make up 85% of the matter in the universe
- Dark matter detector DAMA/LIBRA in Italy
- Earth moves with the Sun for half the year, and in the opposite direction of the Sun for the other half. This could explain DAMA/LIBRA's detected peaks in June and valleys in November
- To see if this theory holds true, an identical detector was built in Australia, where seasons are opposite that of Italian seasons
- Bullet cluster mass imbalance --> dark matter passed right through, while hot gas collided in the center
- WIMP (Weakly interactive massive particles) --> another potential theory w/ same mass as proton
- Sensitive particle detectors must be located deep underground to prevent detector noise
- Idea of a "dark standard model"
- BRO why would you end the video there
- ★ What happened with the sister detector in Australia?

- ★ Is there anything to substantiate the idea of a dark standard model, or is that purely theoretical? Would that also act as a foil to matter in the way that antimatter does, or is there no parallel?
- ★ What is the proffered explanation for the results of DAMA/LIBRA?
- ★ What further experiments are currently being conducted to settle this controversy and answer questions on dark matter?

What Is Dark Matter? An Astrophysicist Explains | Edge Of Knowledge | Ars Technica

- Dark matter is cold (moves slowly compared to the speed of light) and collision-less (doesn't interact with itself or normal matter)
- Stars on the outside edges of galaxies orbit much faster than they should considering only luminous matter
- Neutrinos are a viable example of dark matter, as they don't interact with light, but they're not heavy enough to account for the total mass of dark matter
- Matter is preferred over antimatter--> dark matter might explain baryogenesis
- ★ Are neutrinos actually considered dark matter? Even though they do interact with matter through nuclear force (albeit weakly) + gravitationally?
- ★ There are detectors set up across the globe with the goal of detecting dark matter, with no confirmed findings as of yet. How do we know that these models would be effective if they haven't found anything yet? Are dark matter particles really that rare?

Dark Matter: The Math Behind Galaxy Rotation Curves

- Objects with a larger orbital radius should orbit with a slower velocity, following Newtonian mechanics--> Dark matter causes this relationship to flatten out when plotted
- Gas and dust do not account for enough mass to explain this mathematical relationship
- ★ If the orbital velocity/ radius plot flattens out as radius increases, does this mean that dark matter is concentrated on the outside of galaxies? If so, why?

Dark Matter Halo – Wikipedia

- Dark matter halo – hypothetical region that has decoupled from cosmic expansion and contains gravitationally bound matter (virialized clumps of dark matter, known as subhalos, are bound together by gravity)
- Virialized systems have reached a stable equilibrium (PE and KE are balanced, so no rapid collapsing or expanding is occurring)
- Dark matter halo of a galaxy envelops the galactic disc and extends well beyond the edge of the visible galaxy
- Early galactic formation was made possible by dark matter despite the high temperature of baryonic matter at this time
- ★ What causes decoupling of galaxies and what does that entail?

- ★ What role does the Cosmic Microwave Background play in enhancing our understanding of early galactic formation?

[Astronomical Coordinate Systems – Wikipedia](#)

- Used to describe positions of celestial objects in a 3D space
- Spherical coordinates are projected on the celestial sphere
- Rectangular coordinates have the same fundamental plane and primary direction
- Coordinate systems differ in their choice of fundamental plane
- Horizontal system – based on position of observer on Earth
- Equatorial system – centered at Earth's center, but fixed relative to celestial poles (popular choice for astronomers)
- Ecliptic system – fundamental plane of the Earth's orbit
- Galactic system – uses approximate plane of Milky Way
- Supergalactic system – corresponds to a fundamental plane containing a higher than average number of local galaxies in the sky as seen from Earth
- ★ Chat what is even happening with that last one